ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Technology (CSE)

QUIZ #1 (SET-B) SUMMER SEMESTER, 2023-2024

DURATION: 30 MINUTES FULL MARKS: 15

CSE 4711: Artificial Intelligence

Programmable calculators are not allowed.

Answer **all 2 (two)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

Student ID:

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1. Two friends, Donald and Joseph, are in a fully-observable room that can be represented as an $M \times N$ grid. They want to paint the whole grid and they can start in any two squares of the grid. In each time step, they can either move (UP, DOWN, LEFT, and RIGHT), choose to STAY in the same square, or PAINT the square they are standing on. The actions of Donald and Joseph occur simultaneously. They want to devise the most efficient way to paint the whole room in the shortest possible moves. Note that they cannot leave the grid and both cannot stand on the same square. 6 (CO1) (PO1)

   Formulate the problem as a single-agent search by identifying the state space, start state, actions, successor function, goal test, and cost.

Solution:

- **State space:** Each state consists of the location of Donald and Joseph $((X_D, Y_D), (X_J, Y_J))$ and a 2D array of booleans C (of size $M \times N$) to denote which square has been colored.
- **Start state:** Donald and Joseph can start from any square in the grid and $C_{i,j} = 0$ for $\forall_{i,j} \in M, N$.
- **Actions:** Up, Down, Left, Right, Stay, and Paint for each of them
- **Successor function:** Takes the current state and a valid action to then possibly change the co-ordinates of each person or color a square.
- **Goal test:** Check if $C_{i,j} = 1$ for $\forall_{i,j} \in M, N$.
- **Cost:** Number of moves made.

Rubric:

- 1 point for each correctly answered segment.

2. a) Among the following graph search algorithms, identify the ones that are guaranteed to find the most efficient path for the problem formulated in Question 1: 5 (CO1) (PO1)
   1. Breadth-First Search
   2. Uniform Cost Search
   3. A* Search (with an admissible but not consistent heuristic)

4. A* Search (with a consistent heuristic)
5. A* Search (with a heuristic that returns zero for each state)

Solution:
Algorithms: 1, 2, 4, 5
Rubric:

- 1 point for each correct identification.
- 1 point for each correct omission.

b) Propose a non-trivial admissible heuristic for the problem defined in Question 1. 4 (CO1) (PO1)

Solution:
Since two agents can each paint one square per time step, the minimum number of steps needed to paint all remaining cells is: $h(n) = \frac{U}{2}$ where U is the number of unpainted cells at a given state. thus, this could be a possible admissible heuristic.
Rubric:

- 4 points for an admissible heuristic.